

GENESIS 2.5: opt-in OpenCL and CUDA acceleration for the GENESIS/PGENESIS compartmental neural simulator

Karol Chlasta^{a,*}, Grzegorz Marcin Wójcik^b

^a*Kozminski University, ul. Jagiellońska 57/59, 03-301 Warsaw, Poland; WarsawIQ, Al.
Jana Pawła II 43A/37B, 01-001 Warsaw, Poland*

^b*Maria Curie-Skłodowska University, ul. Akademicka 9, 20-033 Lublin, Poland*

Abstract

GENESIS is a long-established compartmental neural simulator. We release GENESIS 2.5, adding opt-in OpenCL and CUDA backends for the Hodgkin–Huxley channel-update path and `hines_tree_eliminate`, a GPU Hines tree-elimination kernel for dendritic trees, leaving every existing model, script and MPI workflow unchanged. On NVIDIA A100 hardware it reaches $57\times$ end-to-end speedup on a 160 000-compartment population, matching the fp64 CPU to $\sim 10^{-7}$ V. On CPU it runs the Vogels–Abbott COBAHH benchmark $1.63 \pm 0.07\times$ faster than CoreNEURON, and on GPU it overtakes the GPU-native Arbor beyond 76 ms of simulated time, its cost per step being 32% lower. The release also fixes seven latent defects and makes $O(n^2)$ model construction linear.

Keywords: GENESIS, PGENESIS, OpenCL, CUDA, GPU acceleration, compartmental neural simulation, Hines solver

Required Metadata

Current code version

Main text

1. Motivation and significance

*Corresponding author.

Email addresses: `karol@chlasta.pl` (Karol Chlasta), `gmwojcik@umcs.edu.pl` (Grzegorz Marcin Wójcik)

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v2.5 (2026-07-25; OpenCL + CUDA-validated, multi-compartment GPU Hines solve)
C2	Permanent GitHub link to code/repository used for this code version	https://github.com/WarsawIQ/genesis-2.5 (archived release: https://doi.org/10.5281/zenodo.21658389)
C3	Legal Code License	GNU General Public License v2 or later (program); GNU Lesser General Public License v2.1 or later (library portions) – see LICENSE
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	C, OpenCL C, CUDA C++ (accelerator backends), MPI (PGENESIS), Python (benchmark analysis/plotting)
C6	Compilation requirements, operating environments & dependencies	Linux x86_64, GCC, GNU make, an OpenCL 1.2+ runtime (tested: ROCm 6.3.1 and Mesa rusticl) or a CUDA 12.x toolkit (tested: CUDA 12.8, sm_80 NVIDIA A100 and sm_86 NVIDIA A40 on the UMCS cluster) for the GPU backends, an MPI implementation for PGENESIS (MPICH/HYDRA or Open MPI tested on the UMCS cluster)
C7	If available Link to developer documentation/manual	README.md and paper/docs/REPLICATION.md in the repository
C8	Support email for questions	karol@chlasta.pl

Table 1: Code metadata (mandatory)

Compartmental simulators remain essential where channel kinetics and dendritic geometry cannot be abstracted away. GENESIS [1] is among the longest-maintained simulators in this class, with PGENESIS ex-

tending it to MPI; both are still used in production workflows, scripted in SLI, independent of newer ecosystems.

Modern workstations ship capable integrated GPUs, but GENESIS has no accelerator backend: every channel-kinetics update runs on the CPU regardless of available hardware, though per-compartment gating integration is embarrassingly parallel. Existing GENESIS/PGENESIS users have no path to accelerator support without abandoning their models and SLI scripts.

Existing accelerated simulators, and what they ask of a user.

Compartmental simulation on GPUs is not new, and the alternatives are mature. NEURON [11] is the most widely used simulator in this class; CoreNEURON [8] is its optimized compute engine, which strips the interpreter and runs the same models on GPUs through OpenACC, reaching large speedups on multi-rank and multi-GPU systems. Arbor [7] was written from scratch for many-core and GPU hardware, with a performance-portable back end and its own model description. For network models of point neurons rather than detailed morphologies, NEST [12] targets large-scale distributed simulation, and Brian 2 [13] generates code from equations, with GPU execution available through GeNN [14].

Each of these is a strong choice for a new model. None is a path for an existing GENESIS model. CoreNEURON accelerates NEURON’s models, not GENESIS’s; Arbor and Brian 2 require the model to be re-expressed in their own descriptions; and a GENESIS user’s investment is typically in SLI scripts, `.p` cell files and channel prototypes accumulated over years. Porting is not a mechanical translation — as this work found, two implementations of the same published benchmark differ enough in channel kinetics and connectivity construction that establishing their equivalence is itself experimental work (Section 3). The gap GENESIS 2.5 addresses is therefore not “fastest simulator” but “acceleration without migration”: the same models and scripts, unchanged, on the hardware the user already has.

Contributions. This work contributes:

- 1.1 Opt-in OpenCL and CUDA backends for the `hsolve` channel-update path, behind one entry point, leaving every existing CPU, MPI, model and SLI-script workflow unchanged.
- 1.2 `hines_tree_eliminate`, a GPU tridiagonal (Hines [10]) elimination kernel extending acceleration from isopotential cells to axially-coupled dendritic trees, running the identical elimination the CPU solver does.

- 1.3 Seven defect fixes: four in the accelerator, which produced silently wrong results, and three in the Hines solver, which suppressed `inject`-driven transients in any multi-neuron `hsolve`. The latter affect existing users regardless of GPU use.
- 1.4 A latent $O(n^2)$ model-construction bottleneck, unrelated to accelerator support, reduced to linear.
- 1.5 A head-to-head evaluation against NEURON and CoreNEURON on a published network model, in which we verified that the two implementations are equivalent, and timed both arms the same way.

GENESIS 2.5 closes this gap: GENESIS 2.4/PGENESIS 2.4 plus an OpenCL backend for the `hsolve` channel-update path, and a CUDA backend on the same interface (a faithful fp32 port, validated on UMCS cluster NVIDIA A100/A40 nodes; Section 3). No existing CPU, MPI, model, or SLI-script workflow needs to change. While validating this backend we additionally found and fixed three independent solver defects in `hines_solve.c/hines_init.c` affecting any multi-neuron `hsolve` configuration: current injection (`inject`) silently failed to produce a voltage response prior to the fix, which would have invalidated any dynamics-comparing benchmark and may have affected unrelated multi-neuron `hsolve` models elsewhere.

Why a GPU helps here at all is specific to the workload. In a Hodgkin–Huxley model the per-compartment gating-variable integration is the dominant cost, and it is embarrassingly parallel: each compartment’s channel state advances from its own voltage and its own table lookups, with no coupling to any other compartment within the step. The state involved is small and the arithmetic intensity low, so the work maps onto thousands of lightweight threads without the memory pressure that limits many scientific kernels. That makes the channel-update path the natural first target, ahead of the tridiagonal solve.

We provide both OpenCL and CUDA rather than choosing one because they buy different things. OpenCL is vendor-neutral and runs on the integrated GPUs now shipped with ordinary workstations, including devices without double-precision support, which is why the kernel is written in fp32 throughout. CUDA targets the NVIDIA datacenter and desktop cards where most cluster time is available and where the toolchain and profiling tools are standard. Both sit behind one entry point, so a model and its SLI script are unchanged between them and the choice is a build flag.

We name this release “2.5” rather than “3.0” deliberately: “GENESIS 3” [2] produced independent successor lines (Neurospaces/Heccer [3],

MOOSE) evolving into the CBI federation [4, 5, 6], not a single artifact comparable to GENESIS 2.4. GENESIS 2.5 is not a re-architecture and targets existing GENESIS 2.4 deployments that should not need to migrate models or scripts to gain accelerator support.

2. Software description

2.1 Software architecture

GENESIS 2.5 leaves the existing architecture (SLI interpreter, **hsolve** solver, PGENESIS MPI partitioning) unmodified and adds one component: an OpenCL backend invoked from **hsolve** when an attached element uses **chanmode=4** (or 5) with real ion-channel state. Two dispatch modes are implemented:

- *Per-step dispatch* (**ocl_chip_update**): one **clEnqueueNDRangeKernel** call per simulation step, with channel state uploaded and downloaded every step.
- *Multiloop dispatch* (**chip_channel_multiloop**, via **GENESIS_OCL_MULTILOOP=<K>**): a single kernel dispatch runs all K steps in an inner GPU-side loop, amortizing the per-step PCIe transfer cost that otherwise dominates wall time (Section 3).

Every accelerator run emits a non-empty profiling summary, which guards against two silent failures: an **hsolve** with no attached channels never reaching the kernel, and a kernel that fails to build falling back to CPU unnoticed. The kernel is fp32 throughout, so it runs on runtimes without double-precision support; the rest of **hsolve** stays **double**, converting only at the buffer boundary.

Both dispatch modes update each compartment independently, exact only for single-compartment (isopotential) neurons: a real dendritic tree’s compartments are coupled through axial resistance, solved on the CPU by **hsolve** as a compiled bytecode program performing a bottom-up/top-down Gaussian elimination over the tree’s tridiagonal system once per step. We add a third kernel, **hines_tree_eliminate** (**ocl_channel.cl**; CUDA port in **cuda/cuda_channel.cuh**), executing this identical bytecode on the GPU, one thread per disconnected tree (neuron), operating on the same flat opcode/coefficient arrays the CPU solver already builds at **SETUP** time. Because independent trees never reference each other’s rows (the combined program is block-diagonal), threads need no synchronization. The channel and elimination kernels dispatch back to back on the same queue/stream, so the K -step batch still uploads and downloads state once. **ocl_hsolve.c/cuda_hsolve.c** select this path whenever the **hsolve** contains

a multi-compartment tree, leaving the original kernel for single-compartment networks.

2.2 Software functionalities

- Unmodified serial (`genesis`), headless (`nxgenesis`), and MPI (PGENESIS) execution paths.
- OpenCL channel-update kernel for Hodgkin–Huxley-style `tabchannel` gating ($\text{Na } X^3Y^1$, $\text{K } X^4$), dispatched per-step or batched via multiloop.
- A CUDA backend at the same call site (`genesis/src/hines/cuda/`): a line-for-line fp32 port behind an identical entry point, so one model and harness drive either backend.
- Device-side dispatch confirmation distinguishing a genuine GPU run from CPU fallback.
- A driven-spiking benchmark (`hh_spiking_benchmark.g`) exercising the repaired `inject` path.
- Three corrected Hines-solver defects (forward-elimination fast-path guard, backward-elimination root handling, `SET_DIAG/SKIP_DIAG` row selection), independent of GPU support.
- `hines_tree_eliminate`, a GPU tridiagonal elimination kernel in both backends, extending acceleration to axially-coupled dendritic trees (Section 3).
- Five corrected $O(n^2)$ model-construction defects, restoring linear-time construction independent of GPU support (Section 3).

The tridiagonal (Hines) elimination kernel decides whether any of this reaches realistic morphologies. Accelerating channel updates alone leaves the axial coupling between compartments on the CPU, so a multi-compartment model gains little: the solve becomes the bottleneck as soon as the channel work is removed from it. `hines_tree_eliminate` moves that solve onto the device as well, running the identical elimination the CPU performs, one thread per neuron. The consequence is a change in kind rather than degree — acceleration stops being limited to isopotential point-neuron networks and applies to populations of real dendritic trees, which is the regime Section 3 measures.

3. Illustrative examples

Reproducing the CPU baseline. `nxgenesis` on the bundled `Ca18.g` and `Ca17difshell.g` models gives stable, low-variance timings (CV $\approx 1\%$, 30 replicates), confirming ordinary GENESIS workflows are unaffected by this release.

Reproducing the GPU multiloop result. `genesis/Scripts/benchmark/hh_spiking_benchmark.g` builds N driven, spiking Hodgkin–Huxley neurons under one `hsolve (chanmode=4)`; `GENESIS_CUDA_MULTILOOP=K` (or its OpenCL equivalent) dispatches all K steps in one GPU call. Both arms are wall-clocked identically at $K = 50\,000$ on the UMCS A40 and A100.

Table 2 reports both backends after the scheduler and solver fixes. Speedup grows with N throughout, reaching $21.0 \pm 0.3\times$ (CUDA) and $19.2\times$ (OpenCL) on the A40 and $21.9 \pm 0.3\times$ and $20.4\times$ on the A100 at $N = 50\,000$.

Numerical fidelity of the fp32 accelerator. `genesis/Scripts/benchmark/hh1952_ap_verify.g` compares the classic 1952 squid AP across CPU fp64, GPU per-step, and GPU multiloop paths: fp32 agrees with fp64 to $\sim 2 \times 10^{-7}$ V over a single AP, all N neurons identical to $< 10^{-6}$ V (Fig. 1). Over long spiking runs, fp32/fp64 traces drift in spike *phase* but preserve waveform and firing rate, why timing uses driven neurons while correctness uses a single-AP window.

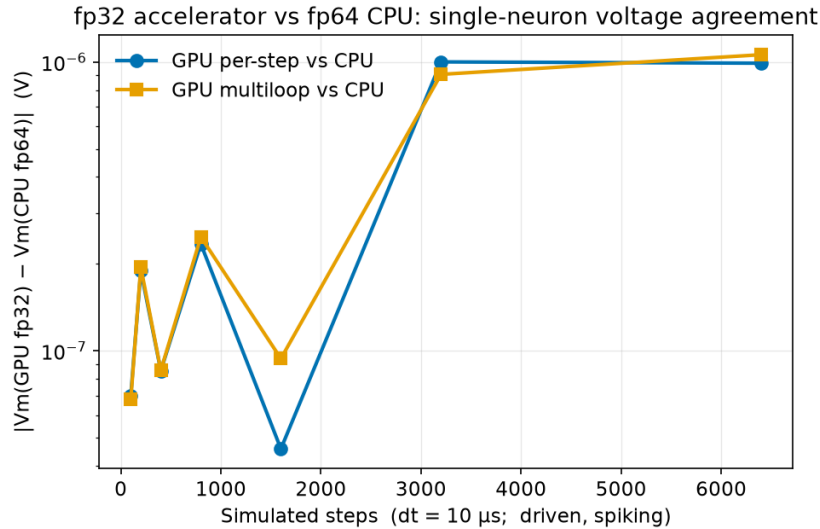


Figure 1: fp32 accelerator vs. fp64 CPU: absolute membrane-voltage difference for a single neuron as a function of simulated steps, for both the per-step and multiloop GPU paths.

CUDA backend on NVIDIA hardware. The CUDA backend builds from the same source (`make USE_CUDA=1`) against the identical harness, and reproduces the fp64 CPU action potential to 7×10^{-8} V, confirming parity with both the CPU and OpenCL paths. Table 2 reports a 10-replicate sweep on the two UMCS cluster GPUs.

These are *end-to-end* speedups: both arms are wall-clocked around the whole process, so construction, transfers and per-step host work are inside the measurement, and every figure in this paper uses that one definition. Timing the two arms by different clocks inflates the ratio by an order of magnitude and can invent apparent differences between cards; measured consistently the A40 and A100 are indistinguishable here ($21.0 \pm 0.3\times$ vs. $21.9 \pm 0.3\times$).

At $N = 500$ the accelerator barely pays for itself: at that size the fixed costs of construction and transfer dominate a run the CPU finishes in under a second.

N	NVIDIA A40		NVIDIA A100	
	CUDA	OpenCL	CUDA	OpenCL
500	$1.5 \pm 0.1\times$	$1.5\times$	$1.8 \pm 0.3\times$	$1.7\times$
5 000	$8.7 \pm 0.3\times$	$7.5\times$	$9.7 \pm 0.4\times$	$8.7\times$
50 000	$21.0 \pm 0.3\times$	$19.2\times$	$21.9 \pm 0.3\times$	$20.4\times$

Table 2: Both accelerator backends (fp32) against the fp64 CPU solver, single-compartment multiloop kernel (`hh_spiking_benchmark.g`, $K = 50\,000$ steps) on the UMCS cluster. End-to-end speedups: both arms of every ratio are wall-clocked around the whole process, so construction, transfers and per-step host work are inside each figure. CUDA is the mean of 10 replicates with propagated uncertainties; OpenCL is the mean of 3. The two cards are close at every size, and OpenCL trails CUDA by roughly 8%, about what a faithful port of the same kernels costs. The OpenCL arm agrees with the fp64 CPU to 3.1×10^{-8} V. Raw data: `cluster_bringup/logs/`.

Multi-compartment GPU acceleration. The multiloop kernel above is exact only for single-compartment (isopotential) neurons: for a real dendritic tree it silently drops all axial coupling. `hines_tree_eliminate` (Section 2.1) performs this elimination on-device, dispatched whenever the `hsolve` contains a multi-compartment tree.

Correctness. Linear-chain and branching benchmarks compare GPU against fp64 CPU across `NCOMP` = 1–32 and up to $N = 10\,000$ neurons: agreement is $\sim 10^{-7}$ V, matching single-compartment fidelity. One topology is not handled: a single-compartment branch off a branch point, which violates the kernel bootstrap’s seed-row assumption. It had been documented but not enforced, so such a model was accelerated and returned wrong voltages silently; both backends now detect it at initialisation and compute that `hsolve` on the CPU.

Speed. Table 3 and Fig. 2 report speedup (mean $\pm$ std, 10 replicates) on the two UMCS cards, sweeping $N = 100$ – $50\,000$ at `NCOMP` = 16, a range unreachable before the model-construction fix (Impact): $N = 50\,000$ took 23.8 ± 0.3 s to build there, versus not finishing before. Branching

topology gives statistically indistinguishable speedups and is omitted from the figure. Dispatch uses $K = 200$ steps ($K = 100$ for $N \geq 5000$), smaller than the single-compartment sweeps’ $K = 50\,000$.

Two figures per card answer different questions. The *step-phase* speedup times the simulation loop alone and measures what the kernel can do: both cards are slower than CPU at $N \leq 500$, pull ahead past $N \approx 500$ – 1000 , and reach $37.3 \pm 0.1 \times$ (A40) and $80.8 \pm 1.0 \times$ (A100) at $N = 50\,000$, neither plateaued. The *end-to-end* speedup wall-clocks the whole process, which is what a user waits for: it is far lower, $3.62 \pm 0.05 \times$ (A40) and $3.90 \pm 0.07 \times$ (A100) at $N = 50\,000$. The gap is not a kernel deficiency but Amdahl’s law applied to a short run — at $K = 200$ steps the unaccelerated construction phase dominates, and the end-to-end figure climbs toward the step-phase ceiling as K grows. Quoting only the step-phase number would overstate what the accelerator delivers in practice.

The cards nearly coincide end-to-end despite the A100 leading by more than $2 \times$ on the step phase — consistent with an fp32 kernel using no tensor cores, the GPU phases taking 7.83s (A40) against 8.06s (A100) at $N = 50\,000$. Raw data: `cluster_bringup/logs/weekend_campaign*_20260816_clean.csv` and `..._multicomp_walltime_createmap*_clean.csv`.

Parallelization granularity. `hines_tree_eliminate` assigns one GPU thread per tree, so speedup comes from spreading many neurons across threads, not from parallelizing elimination *within* a tree. For a single detailed morphology alone ($N = 1$), only one thread is active: measured directly, the kernel is ~ 20 – $21 \times$ *slower* than the CPU at `NCOMP` = 2000 and 8000, stable with tree size as expected for a single-thread bottleneck. The kernel therefore accelerates population-scale simulation (Table 3), not the single-detailed-cell regime.

Population-scale model construction was $O(n^2)$; now $O(n)$. Pushing toward the $\sim 31\,000$ -neuron Blue Brain Project scale [9] exposed a defect independent of GPU support: 527 000 compartments did not finish building within 900s. Profiling gave a fitted exponent of 2.31, traced to five pre-existing linear-scan-per-element patterns in element creation, path resolution and `hsolve` SETUP, none GPU-specific — each rescanning siblings, paths or compartments where hashing or amortized growth would do. Fixing all five (Table 4, Fig. 3) gives an exponent of 0.99: that population now builds in 14.6 ± 0.2 s, and 1.7 million compartments in 51.5 ± 0.6 s. Output is byte-identical on every validation script, confirming a pure performance fix; full diagnosis in `genesis/src/hines/GPU_HINES_SOLVE_DESIGN.md`.

		step-phase		end-to-end	
N	total comps	A40	A100	A40	A100
100	1 600	0.2 ± 0.02	0.3 ± 0.02	0.43 ± 0.05	0.61 ± 0.02
500	8 000	1.1 ± 0.03	1.7 ± 0.11	0.77 ± 0.02	1.04 ± 0.03
1 000	16 000	2.0 ± 0.08	3.1 ± 0.07	1.10 ± 0.08	1.46 ± 0.04
2 000	32 000	4.5 ± 0.39	6.4 ± 0.18	1.37 ± 0.06	1.57 ± 0.05
3 000	48 000	6.7 ± 0.03	12.0 ± 0.37	1.87 ± 0.07	1.89 ± 0.04
5 000	80 000	6.9 ± 0.04	14.5 ± 0.47	2.60 ± 0.06	2.66 ± 0.03
7 000	112 000	9.8 ± 0.05	22.0 ± 0.43	2.92 ± 0.05	3.08 ± 0.03
10 000	160 000	13.7 ± 0.06	30.6 ± 0.68	3.01 ± 0.04	3.23 ± 0.04
15 000	240 000	20.3 ± 0.12	42.3 ± 0.77	3.27 ± 0.05	3.52 ± 0.03
20 000	320 000	25.3 ± 0.12	51.6 ± 1.16	3.39 ± 0.04	3.71 ± 0.04
30 000	480 000	31.9 ± 0.13	66.1 ± 1.12	3.53 ± 0.05	3.87 ± 0.05
40 000	640 000	35.2 ± 0.13	75.5 ± 0.96	3.56 ± 0.04	3.90 ± 0.04
50 000	800 000	37.3 ± 0.07	80.8 ± 1.01	3.62 ± 0.05	3.90 ± 0.07

Table 3: Multi-compartment GPU tree-elimination (fp32) vs. CPU (fp64), UMCS A40/A100, NCOMP = 16, linear-chain, batched multiloop ($K = 200$, $K = 100$ for $N \geq 5000$); mean $\pm$ std over 10 replicates. *Step-phase* times the simulation loop alone and measures kernel capability; *end-to-end* wall-clocks the whole process, including model construction. The two arms of the end-to-end measurement are timed identically – see `cluster_bringup/54_multicomp_walltime.sh`. The end-to-end runs build the population with a single `createmap` from a prototype, as network models do, rather than the explicit SLI loop of the original benchmark; the step-phase figures are unaffected by that choice, since they exclude construction. “–” = not measured. Raw data: `cluster_bringup/logs/`.

N	total comps	before (s)	after: mean $\pm$ std (s)	CV
1 000	17 000	0.82	0.57 ± 0.08	13.2%
2 000	34 000	2.03	0.93 ± 0.003	0.3%
4 000	68 000	8.24	1.83 ± 0.04	1.9%
8 000	136 000	105.45	4.08 ± 0.03	0.7%
16 000	272 000	did not finish (60s+)	7.54 ± 0.02	0.2%
31 000	527 000	did not finish (900s+)	14.57 ± 0.18	1.2%
50 000	850 000	–	23.80 ± 0.32	1.3%
70 000	1 190 000	–	34.25 ± 0.44	1.3%
100 000	1 700 000	–	51.48 ± 0.61	1.2%

Table 4: Model-construction wall time (construction + `hsolve SETUP` + one step), `hh_branching_multicompartment_benchmark.g`, local machine, before vs. after the five fixes. “After”: mean $\pm$ std, 3 replicates; “before”: single replicate, not re-measured beyond $N = 8000$ (“–” = never attempted). Fitted exponent: 2.31 before, 0.99 after. Raw data: `experiments/data/construction_scaling_before_after.csv`.

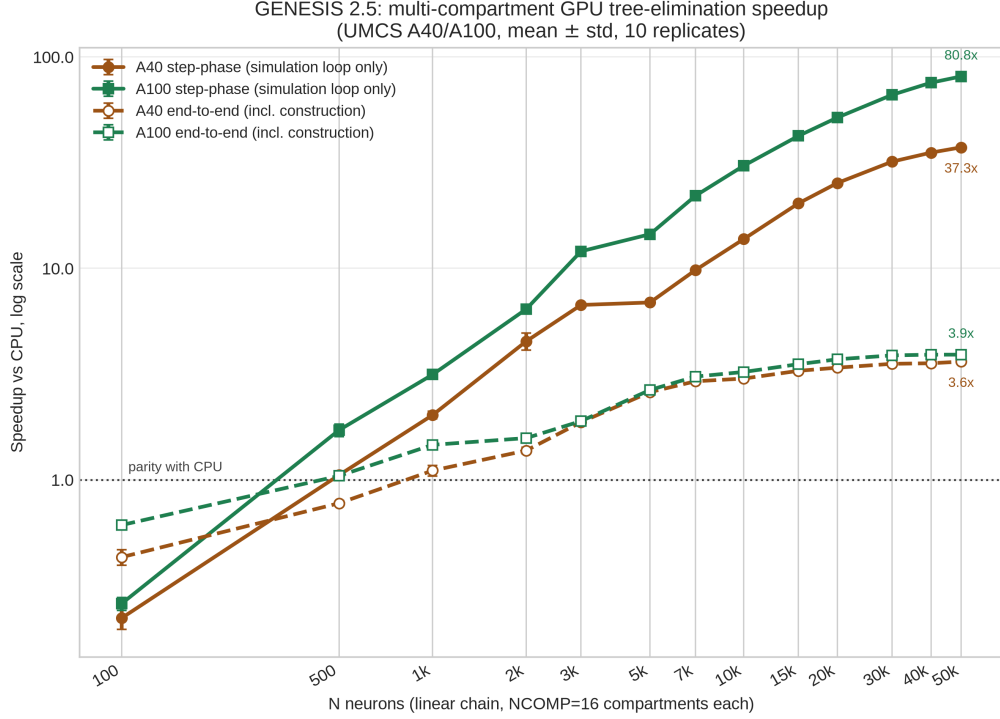


Figure 2: Multi-compartment GPU tree-elimination speedup vs. CPU (mean $\pm$ std, 10 replicates; log-log), UMCS A40/A100, sweeping N at $NCOMP = 16$ (linear-chain). Colour identifies the card, line style the metric: solid = step-phase (simulation loop only), dashed = end-to-end (whole process, including model construction). Both cards start below CPU parity and cross it near $N \approx 500$ – 1000 . The step-phase curves keep climbing to $80.8\times$ (A100) and $37.3\times$ (A40) without plateauing, but the end-to-end curves flatten near $3.9\times$ and $3.6\times$: at $K = 200$ steps the unaccelerated construction phase bounds the achievable speedup, and only the step-phase curves measure the kernel itself. The two cards nearly coincide end-to-end despite the A100’s large step-phase lead, as expected for an fp32 kernel using no tensor cores.

Speedup grows with simulation length. Model construction is a fixed one-time cost, so end-to-end speedup depends on run length. Sweeping K at fixed $N = 5000$ shows end-to-end speedup rising toward the step-phase ceiling, so the accelerator pays off most for the long runs typical of real modelling studies (`experiments/data/campaign_ksweep.csv`).

The multi-compartment figures are a floor set by the benchmark, not a ceiling set by the kernel. The sweeps above use $K = 200$ ($K = 100$ for $N \geq 5000$), which at $dt = 50\mu s$ is 5–10 ms of simulated activity, short enough that fixed costs dominate both arms. Repeating the measurement at $N = 10\,000$ over a range of K (Table 5)

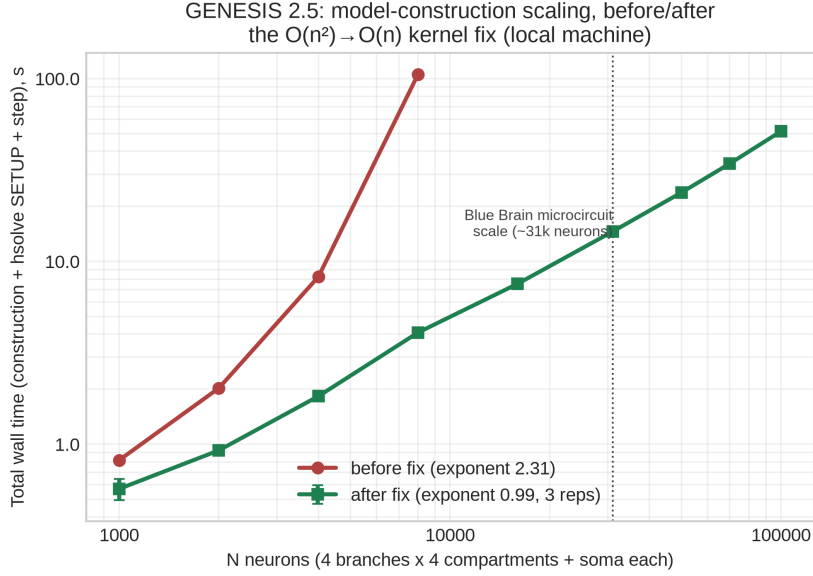


Figure 3: Model-construction wall time vs. N , log-log, before (single replicate) and after (mean $\pm$ std, 3 replicates) the five $O(n^2)$ fixes, extended to $N = 100\,000$. The “before” series stops at $N = 8000$: larger N did not finish within the tested budget.

separates them: end-to-end speedup rises from $4.3\times$ at $K = 200$ to $57.0 \pm 0.9\times$ at $K = 5000$, and step-phase from $13.7\times$ to $116.2\times$. Both grow because the one-time costs they contain are amortized over more steps: model construction for the end-to-end figure, buffer setup and the single batched dispatch for the step-phase figure.

Fitting fixed and marginal costs separates them. Construction takes ≈ 1.3 s on both arms, and one step costs 2.31×10^{-2} s on the CPU against 1.42×10^{-4} s on the GPU, a ratio of $163\times$, with the device profile agreeing independently at $140.4\,\mu\text{s}$ per step. At $K = 5000$ the GPU still reproduces the CPU voltages to $\sim 10^{-7}$ V, so this is amortization, not skipped work. Table 3 reports the short- K figures because they are what the standing benchmark measures, but a 250 ms simulation of this population already reaches $57\times$ end-to-end.

Reproducing the PGENESIS scaling result. The existing MPI path is unaffected; a strong-scaling sweep ($N = 2400$ HH neurons, $P = 1\text{--}24$ ranks, 12-core/24-thread host) gives near-ideal scaling at $P = 2$ ($E_P = 0.95$), an efficiency knee near $P \approx 5$, and a recommended $P = 4\text{--}8$ ($3.1\text{--}4.0\times$, 50–78% efficiency) for this problem size.

Head-to-head against NEURON and CoreNEURON. The speedups above are internal (GPU vs. CPU within GENESIS) and say nothing about how GENESIS compares with other simulators. We there-

<i>K</i> steps	CPU (s)	GPU (s)	End-to-end
200	5.86	1.37	4.28 $\pm$ 0.12
500	13.00	1.42	9.18 $\pm$ 0.25
1 000	24.87	1.49	16.73 $\pm$ 0.36
2 000	48.03	1.63	29.48 $\pm$ 0.91
5 000	116.75	2.05	57.01 $\pm$ 0.88

Table 5: End-to-end wall clock against simulation length, $N = 10\,000$ neurons $\times$ 16 compartments (160 000 compartments), UMCS A40, mean $\pm$ std over 5 replicates, both arms timed identically. The fixed cost each arm carries, ≈ 1.3 s of model construction, is unchanged across the sweep, so the speedup rises as K grows: at $K = 200$ construction is most of the GPU run, at $K = 5000$ it is under a tenth of it. Raw data: `cluster_bringup/logs/multicomp_ksweep_inf02_A40_20260816.csv`, produced by `cluster_bringup/55_multicomp_ksweep.sh`.

fore ran the Vogels–Abbott COBAHH network [15, 16] in GENESIS (Scripts/VAnet2) and in the reference NEURON implementation (ModelDB 83319, NEURON/cobahh) on the same cluster node, all arms single-threaded on CPU (Table 6, Fig. 4). GENESIS 2.5 completes the 5.0 s simulation in 46.9 ± 2.1 s, $1.63 \pm 0.07\times$ faster than CoreNEURON and $2.04 \pm 0.09\times$ faster than NEURON 9.0.2. GENESIS sustains 8.5×10^6 neuron-steps/s against 5.2×10^6 for CoreNEURON and 4.2×10^6 for NEURON.

CoreNEURON exists to make NEURON models faster, discarding the interpreter and rebuilding the data layout for vectorisation, while the GENESIS arm here uses no accelerator at all. CoreNEURON is in turn $1.25\times$ faster than NEURON 9.0.2, the expected ordering, which suggests that arm is configured correctly.

Because these are independent implementations, we verified their equivalence: both have 4000 single-compartment cells with 80 synapses each, both are driven externally for the first 50 ms only, both use $dt = 0.05$ ms for 5.0 s, and both settle into the same regime (26.8 against 27.9 Hz), so neither does materially less synaptic work. Obtaining the GENESIS rate required instrumenting every spikegen: the benchmark records three cells, two of them underconnected grid-edge cells.

The stock benchmark cannot run under CoreNEURON at all, since its channels are ChannelBuilder objects rebuilt inside the interpreter. We transcribed them into NMODL, reproducing the published COBAHH rates, after which CoreNEURON yields a spike train byte-identical to NEURON’s. Harness and mechanisms: `cluster_bringup/coreneuron/`. **CoreNEURON on the GPU.** CoreNEURON exists to run on accelerators, so a CPU-only comparison would flatter us. Building its

OpenACC backend with NVHPC left NEURON itself unable to start, so we ran CoreNEURON on its own: NEURON writes the model out with `nrncore_write`, and `special-core` reads those files and simulates without it, which keeps NEURON on its working GCC build. On one A100 CoreNEURON completes the same network in 27.0 ± 0.1 s — $2.83\times$ faster than its own CPU arm, and $1.74\times$ faster than GENESIS. Its spike count differs from the CPU arms by 6% (592 865 against 558 824), the same activity regime, so the times remain comparable. This is the boundary of what the present release offers. On a spiking network of single-compartment cells GENESIS 2.5 cannot use its accelerator at all, because VAnet2 places one spikegen per `hsolve` (Section 2.1), so a GPU-enabled CoreNEURON beats our CPU solver by $1.74\times$. Removing that restriction is the main piece of work left (Section 5). Where the accelerator does apply — multi-compartment populations — the comparison below is GPU on both sides.

Simulator	Channels	Wall (s)	Spikes
CoreNEURON 9.0.2, GPU	compiled NMODL	27.0 ± 0.1	592 865
GENESIS 2.5 (VAnet2)	tabulated	46.9 ± 2.1	536 600
CoreNEURON 9.0.2	compiled NMODL	76.5 ± 0.3	558 824
NEURON 9.0.2	compiled NMODL	95.8 ± 0.2	558 824
NEURON 9.0.2	ChannelBuilder	123.3	574 138
NEURON 8.0.2	ChannelBuilder	76.5	—

Table 6: Vogels–Abbott COBAHH network, 4000 single-compartment cells, 80 synapses per cell, $dt = 0.05$ ms, 5.0 s simulated, UMCS node inf03. All arms are single-threaded CPU except the first, which runs on one A100. Mean $\pm$ std over 3 replicates for the top four rows; the two ChannelBuilder rows are single runs, shown for reference only. GENESIS 2.5 is $1.63 \pm 0.07\times$ faster than CoreNEURON on CPU, and $1.74\times$ slower than CoreNEURON on GPU — on this benchmark GENESIS has no accelerator path (Section 2.1). The two simulators reach the same activity regime (26.8 vs. 27.9 Hz), so the wall times reflect comparable work. All NEURON and CoreNEURON replicates returned an identical spike count, confirming the runs are deterministic. NEURON 8.0.2 is included because its pip distribution ships the CoreNEURON Python interface but not the engine, so the CoreNEURON arm requires NEURON 9. Harness: `cluster_bringup/coreneuron/`.

GPU against a GPU-native simulator. The comparison above is CPU on both sides, because the Vogels–Abbott network is spiking and the accelerator declines it (Section 2.1). To test the accelerator across simulators we used a model it supports — $N = 10\,000$ neurons, each a chain of 16 compartments, HH channels, current injection, no synapses — reimplemented with matching geometry, conductances, injection and timestep in NEURON and in Arbor [7], which was written from scratch for GPU hardware.

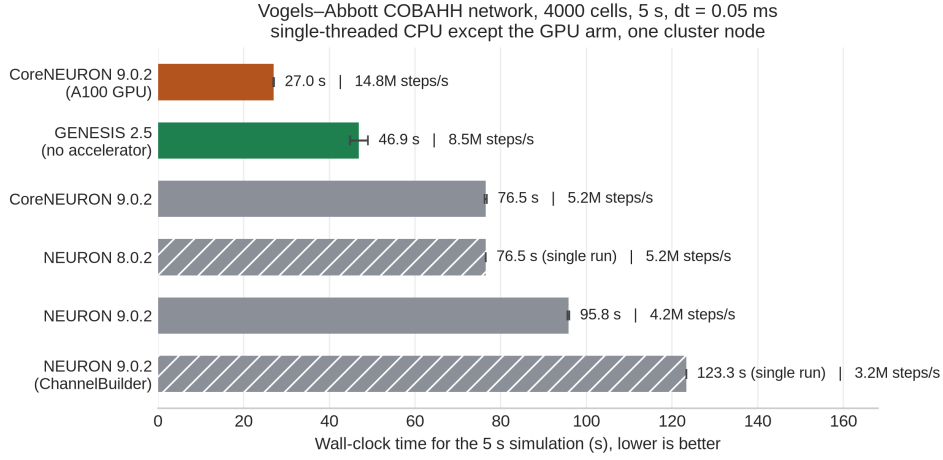


Figure 4: GENESIS 2.5 against NEURON and CoreNEURON on the Vogels–Abbott COBAHH network, one UMCS node. All arms are single-threaded CPU except the top one, CoreNEURON on an A100, coloured apart. Hatched bars are single runs; the others are means of 3 replicates with standard deviations. Throughput is quoted over the 4×10^8 neuron-steps the 5 s simulation advances. Against CoreNEURON on CPU, the GENESIS arm completes the network in two thirds of the time using no accelerator at all; against CoreNEURON on the GPU it is $1.74\times$ slower, because a spiking network of single-compartment cells is the one case this release cannot accelerate.

Which simulator is faster depends on run length, and the crossover is measurable. Over $K = 5000$ steps on one node (Table 7) Arbor finishes in 1.56 ± 0.02 s against our 1.75 ± 0.08 s; over $K = 50\,000$ the order reverses, 4.66 ± 0.07 s against Arbor’s 5.92 ± 0.06 s. Both are linear in K . Fitted over six run lengths from $K = 1000$ to $50\,000$, GENESIS costs 1.30 s plus $65.9 \mu\text{s}$ per step and Arbor 1.07 s plus $96.7 \mu\text{s}$: our marginal cost is 32% lower while theirs starts 0.23 s ahead, so the lines cross at $K \approx 7600$, or 76 ms of simulated time at $dt = 0.01$ ms (Fig. 5). Below it Arbor wins on setup; above it, where modelling studies sit, GENESIS 2.5 is faster. The device profile explains the split: our step phase exceeds kernel time by 0.21 s at both run lengths, so that excess is one-time dispatch setup, not per-step work, and it costs us the short-run case. This is the cost of the design that keeps existing models working: Arbor generates specialised code per mechanism, while our kernels interpret the `hsolve` opcode program at run time. On single-threaded CPU the GENESIS solver is $2.02\times$ slower than CoreNEURON: it is the accelerator, not the solver, that this release contributes. Arbor is also faster on 128 CPU threads than on the GPU at this size, its construction cost dominating.

Simulator	Backend	Wall (s)
<i>K = 5000 steps (50 ms simulated)</i>		
Arbor 0.10.0	CPU, 128 threads	1.09 ± 0.04
Arbor 0.10.0	GPU (A100)	1.56 ± 0.02
GENESIS 2.5	GPU (A100)	1.75 ± 0.08
<i>K = 50 000 steps (500 ms simulated)</i>		
GENESIS 2.5	GPU (A100)	4.66 ± 0.07
Arbor 0.10.0	GPU (A100)	5.92 ± 0.06
CoreNEURON 9.0.2	CPU, 1 thread	60.94 ± 0.15
NEURON 9.0.2	CPU, 1 thread	82.82 ± 2.00
GENESIS 2.5	CPU, 1 thread	123.41 ± 1.75

Table 7: Multi-compartment Hodgkin–Huxley benchmark across simulators: $N = 10\,000$ neurons $\times$ 16 compartments (160 000 compartments), $dt = 0.01$ ms, all on UMCS node inf03 (Xeon Platinum 8358, NVIDIA A100), mean $\pm$ std over 3 replicates, both arms of every pair timed identically. Thread counts differ by simulator and are stated: Arbor uses all cores by default, the others are single-threaded. The two run lengths bracket the crossover at $K \approx 7600$. The models match in geometry, passive properties, conductances, injection and timestep; this is a throughput comparison and not a claim that the three produce identical voltages. Harness and raw data: [cluster_bringup/coreneuron/](https://github.com/cluster-bringup/coreneuron/).

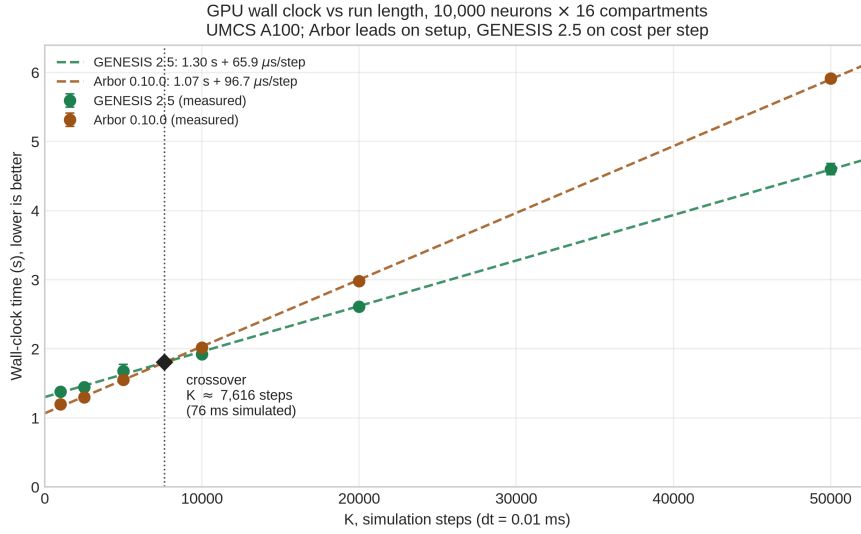


Figure 5: GPU wall clock against run length for the multi-compartment benchmark, $N = 10\,000$ neurons $\times$ 16 compartments on the UMCS A100, six run lengths, mean $\pm$ std over 3 replicates each. Both simulators are linear in K with different intercepts and slopes: Arbor starts 0.23s ahead on setup, GENESIS 2.5 advances each step 32% more cheaply. Neither is faster in general — the lines cross at $K \approx 7600$, and a comparison at a single run length reports whichever side of that the author happened to choose.

A reproduction pack is included: `shreproduce/run_all.sh` re-measures every claim here, regenerates the figures from those measurements, and prints each published value beside the reader’s own. Raw data and full steps: `paper/docs/REPLICATION.md`.

4. Impact

GENESIS/PGENESIS users with channel-kinetics-heavy models now have a path to GPU acceleration (Section 3) without migrating models, SLI scripts, or MPI workflows. That is the principal barrier an architecturally different successor such as MOOSE would otherwise impose. The three Hines-solver defects fixed along the way (Section 2) are independent of GPU support and affect any multi-neuron `hsolve` configuration: before the fix, `inject`-driven voltage transients were silently suppressed for every neuron but the first, with no error raised. That defect predates this release and is now fixed for any multi-neuron `hsolve` user, whether or not they use an accelerator.

A broken $O(N)$ scheduler loop, re-scheduling neurons $2 \dots N$ every step, is fixed; that repair makes Section 3’s throughput attainable.

The accelerator has a deliberate scope: both kernels interpret the `hsolve` opcode program directly and implement the five opcodes a current-driven `tabchannel` model produces. Compartments needing anything else (synaptic channels, a `spike` element, GHK, concentration pools) are detected at `SETUP` and computed on the CPU. Validating this on a published network model exposed a fourth latent defect: the detection existed but its result was never consulted, so such compartments reached a kernel that mis-parses their opcode stream, giving wrong voltages with no error raised. It affects any real network model using those elements. The check is now enforced, and that boundary defines what this release accelerates.

Pushing toward the Blue Brain Project’s scale [9] turned up a defect entirely separate from the GPU work: five pre-existing linear-scan patterns compounding into $O(n^2)$ model-construction time (Section 3). None is accelerator-specific, so fixing all five benefits any GENESIS/PGENESIS user building a large model, whether or not an accelerator is available.

5. Conclusions

GENESIS 2.5 adds opt-in OpenCL and CUDA acceleration to GENESIS 2.4/PGENESIS 2.4 for Hodgkin–Huxley `tabchannel` models, without changes to model files, SLI scripts, or MPI workflows (Section 3 reports speedup and fidelity for both integrated-GPU and cluster-class hardware). Beyond the single-compartment multiloop kernel, `hines_tree_eliminate` extends acceleration to axially-coupled multi-compartment

trees at population scale; the single-detailed-cell regime is not helped by this design, which we measure and report.

The release also carries four correctness fixes independent of accelerator use, benefiting any GENESIS/PGENESIS user. Three are latent Hines-solver defects that silently suppressed `inject`-driven voltage transients in any multi-neuron `hsolve`. The fourth is in the accelerator dispatch itself: compartments using opcodes the kernels do not implement were sent to the GPU anyway, producing silently wrong results; they are now detected and computed on the CPU. Separately, five pre-existing linear-scan patterns compounding into $O(n^2)$ model-construction time are fixed, restoring linear scaling and making population-scale models buildable at all.

Measured against other simulators on a published network model, GENESIS 2.5 is $1.63 \pm 0.07\times$ faster than CoreNEURON and $2.04 \pm 0.09\times$ faster than NEURON 9.0.2 for single-threaded CPU execution of the Vogels–Abbott COBAHH benchmark, after verifying that the two implementations are equivalent (Section 3). That benchmark is a spiking network, which this release cannot accelerate, and CoreNEURON on an A100 runs it $1.74\times$ faster than our CPU solver. On multi-compartment populations, where the accelerator does apply, GENESIS 2.5 overtakes the GPU-native Arbor beyond 76 ms of simulated time.

Future work. Four directions follow directly from the limits measured here, in what we judge to be descending order of value to users.

Scale. The kernel assigns one thread per tree, so it accelerates population-scale simulation and not the single detailed cell, where only one thread is active and the GPU is $\sim 20\times$ slower than the CPU. Intra-tree parallelism, such as cyclic reduction, would extend acceleration to the detailed-morphology regime much of the literature works in. This is where the present design is furthest from complete.

Coverage. Synaptically coupled networks are declined rather than accelerated, since the kernels implement only the opcodes a current-driven `tabchannel` model produces. Extending coverage to synaptic channels, spike generators and concentration pools would reach the models most GENESIS users run. A structural obstacle sits behind it: GENESIS gives each `hsolve` one spike generator, so a spiking network becomes one solver per cell and dispatch overhead dominates. Batching dispatch across `hsolve` elements is the fix, and it interacts with event ordering, since a spike raised inside a batch cannot reach another cell’s synapses before the batch ends.

Comparison. CoreNEURON’s multi-rank MPI configuration, and Arbor on a spiking network, are still unmeasured and would place GEN-

ESIS 2.5 more completely.

Precision. The kernels are fp32 throughout, which lets them run on integrated GPUs lacking double precision at a cost of $\sim 10^{-7}$ V against the fp64 CPU. Whether that divergence stays bounded over the much longer runs used in plasticity studies is not established here.

Packaging into a container-based deployment pipeline [17] is in progress.

Acknowledgements

The authors thank the Maria Curie-Skłodowska University (UMCS) in Lublin and the LubMAN UMCS computing centre for access to the “Lunar” cluster (NVIDIA A100 and A40 GPU nodes) at the UMCS Institute of Mathematics, commissioned in 2015 [18] and since upgraded with the GPU nodes used in this work, and Karol S. Karpowicz and Prof. Przemysław Stpiczyński for provisioning and support of the compute environment on which the CUDA backend was built and validated. The authors also thank WarsawIQ for the AMD Radeon 890M integrated GPU used for the laptop-class benchmarks reported here.

References

- [1] Bower JM, Beeman D. *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural Simulation System*. Springer, 1998.
- [2] GENESIS 3 Development Plans. <http://www.genesis-sim.org/GENESIS/G3/G3-dev-plans.html>
- [3] Cornelis H, Coop AD, Rodriguez AL, Beeman D, Bower JM. GENESIS 3.0 functionality enhanced by interface to multiple types of database. *BMC Neuroscience* 2011, 12(Suppl 1):P15.
- [4] Cornelis H, Edwards M, Coop AD, Bower JM. The CBI architecture for computational simulation of realistic neurons and circuits in the GENESIS 3 software federation. *BMC Neuroscience* 2008, 9(Suppl 1):P88.
- [5] Cornelis H, Lu H, Esquivel A, Bower JM. Modeling a single dendritic compartment using Neurospaces and GENESIS-3. *BMC Neuroscience* 2007, 8(Suppl 2):P3.
- [6] Cornelis H, Bampasakis D, Steuber V, Bower JM. Interoperability in the GENESIS 3.0 Software Federation: the NEURON Simulator as an Example. *BMC Neuroscience* 2013, 14(Suppl 1):P33.

- [7] Akar NA, Cumming B, Karakasis V, Küsters A, Klijn W, Peyser A, Yates S. Arbor – A Morphologically-Detailed Neural Network Simulation Library for Contemporary High-Performance Computing Architectures. In: *27th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP)*, 2019, pp. 274–282. doi:10.1109/EMPDP.2019.8671560.
- [8] Kumbhar P, Hines M, Fouriaux J, Ovcharenko A, King J, Delalondre F, Schürmann F. CoreNEURON: An Optimized Compute Engine for the NEURON Simulator. *Frontiers in Neuroinformatics* 2019, 13:63. doi:10.3389/fninf.2019.00063.
- [9] Markram H, Muller E, Ramaswamy S, et al. Reconstruction and Simulation of Neocortical Microcircuitry. *Cell* 2015, 163(2):456–492. doi:10.1016/j.cell.2015.09.029.
- [10] Hines M. Efficient computation of branched nerve equations. *International Journal of Bio-Medical Computing* 1984, 15(1):69–76. doi:10.1016/0020-7101(84)90008-4.
- [11] Hines ML, Carnevale NT. The NEURON Simulation Environment. *Neural Computation* 1997, 9(6):1179–1209. doi:10.1162/neco.1997.9.6.1179.
- [12] Gewaltig M-O, Diesmann M. NEST (NEural Simulation Tool). *Scholarpedia* 2007, 2(4):1430. doi:10.4249/scholarpedia.1430.
- [13] Stimberg M, Brette R, Goodman DFM. Brian 2, an intuitive and efficient neural simulator. *eLife* 2019, 8:e47314. doi:10.7554/eLife.47314.
- [14] Yavuz E, Turner J, Nowotny T. GeNN: a code generation framework for accelerated brain simulations. *Scientific Reports* 2016, 6:18854. doi:10.1038/srep18854.
- [15] Vogels TP, Abbott LF. Signal Propagation and Logic Gating in Networks of Integrate-and-Fire Neurons. *Journal of Neuroscience* 2005, 25(46):10786–10795. doi:10.1523/JNEUROSCI.3508-05.2005.
- [16] Brette R, Rudolph M, Carnevale T, Hines M, Beeman D, Bower JM, et al. Simulation of networks of spiking neurons: A review of tools and strategies. *Journal of Computational Neuroscience* 2007, 23(3):349–398. doi:10.1007/s10827-007-0038-6.
- [17] Chlasta K, Sochaczewski P, Wójcik GM, Krejtz I. Neural simulation pipeline: Enabling container-based simulations on-premise and

in public clouds. *Frontiers in Neuroinformatics* 2023, 17:1122470. doi:10.3389/fninf.2023.1122470.

- [18] Maria Curie-Skłodowska University (UMCS), Instytut Matematyki. Superkomputer LUNAR uruchomiony w Instytucie Matematyki UMCS. <https://www.umcs.pl/pl/aktualnosci,57,superkomputer-lunar-uruchomiony-w-instytucie-matematyki-umcs,27879.htm>, 2015.

Current executable software version

Not applicable; this release is distributed as source code only (see Current code version above).